



Biochip Course

Sample Preparation

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a.a. 2022/2023

April 17th 2023



Definition

The sample preparation encompasses all the steps necessary to convert a raw sample into a solution ready for detection.

It must (should) be integrated in the biochip (recent focus)

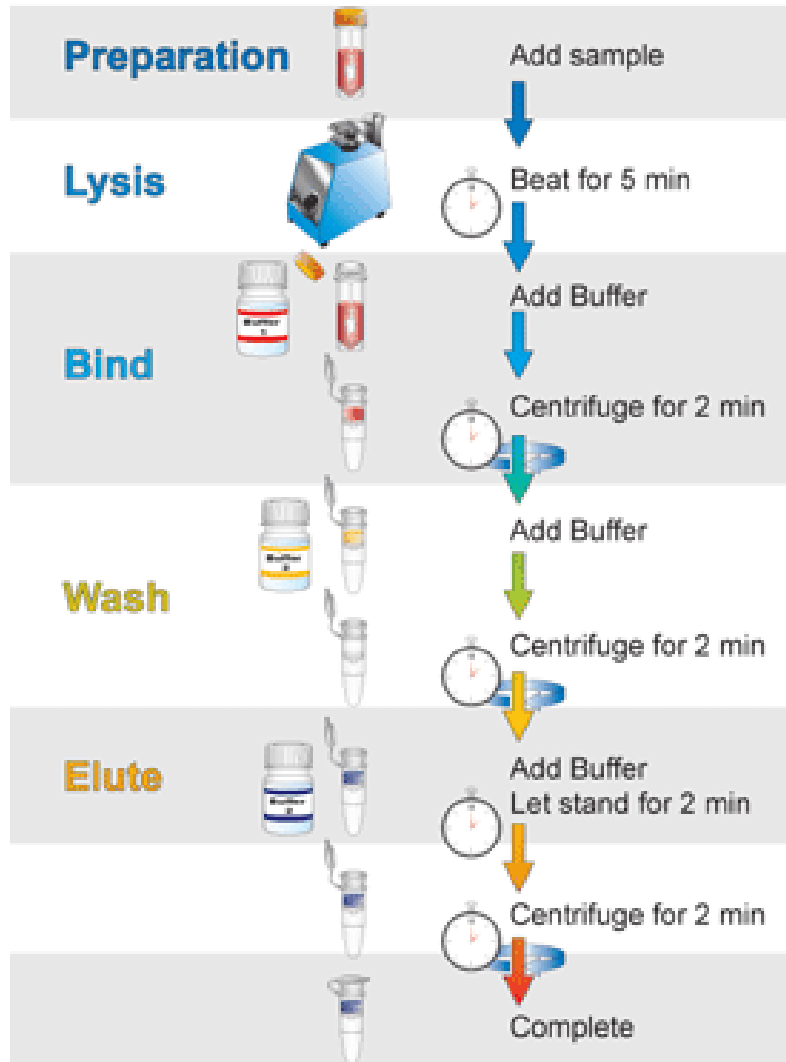
Possible steps:

1. Purification (separation, DEP, H filter...)
2. Concentration (to reach minimum concentration)
3. Extraction (cell *lysis* to extract molecules)
4. Mixing with reagents
5. Labeling with marker (fluorescent or magnetic)
6. Amplification (for DNA, thermal PCR...)

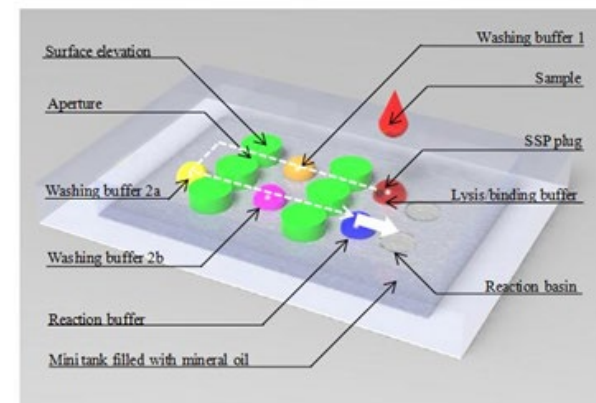
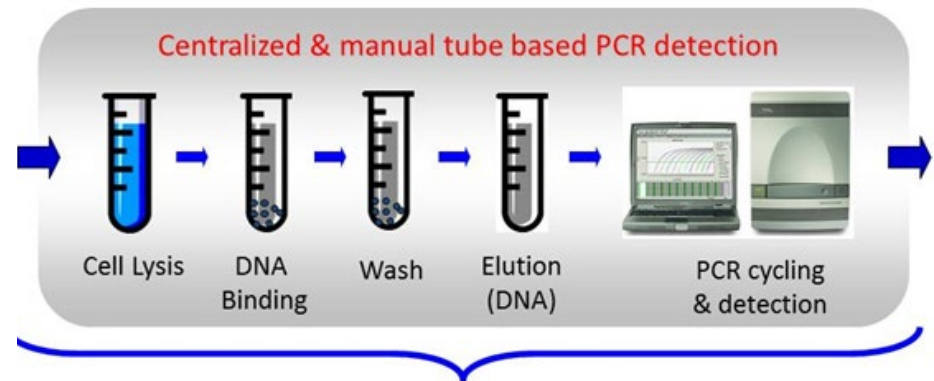


Integration of Sample Preparation Protocols

Purifying DNA



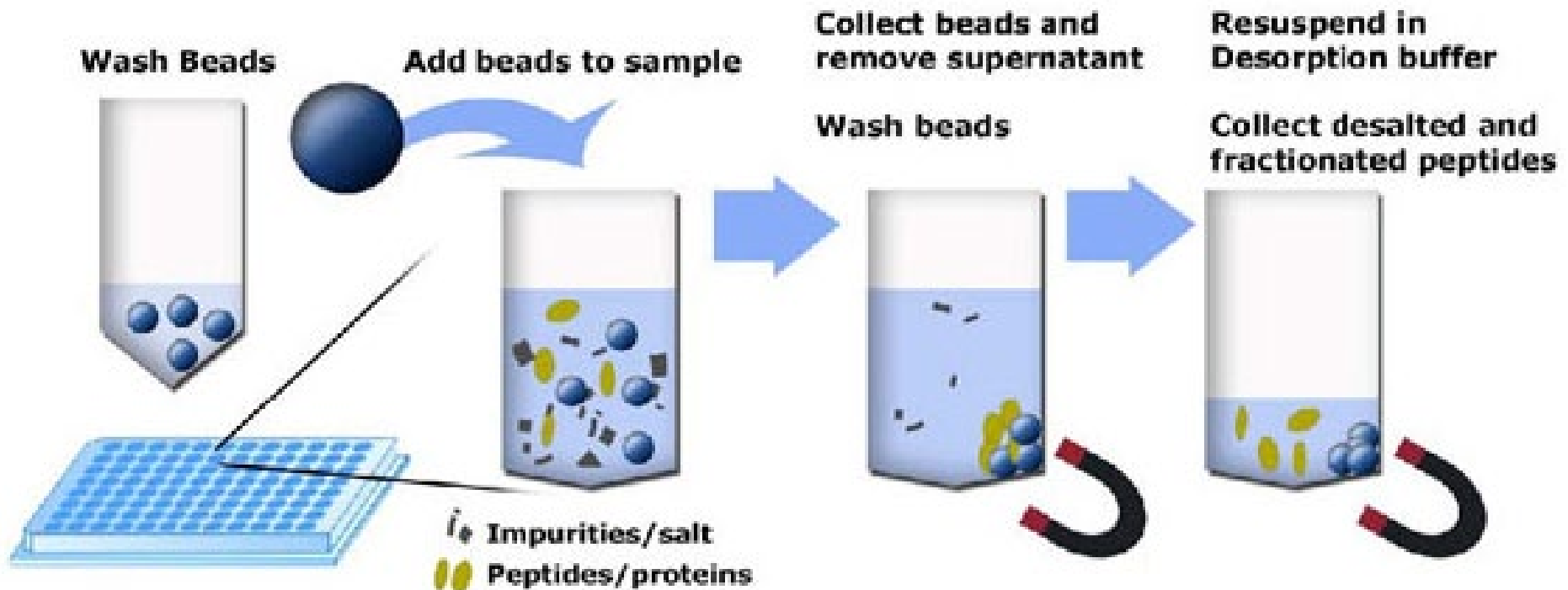
Temporal sequence
becomes controlled
by space





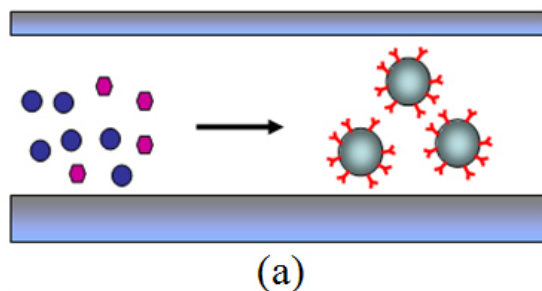
Concentration

Purification and concentration at the same time with magnetic beads functionalized to capture specific molecules

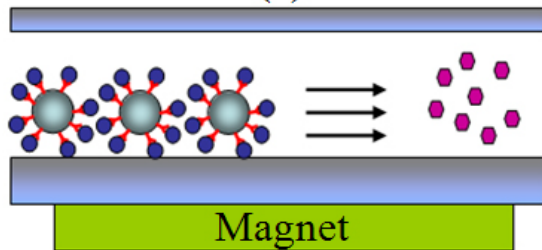




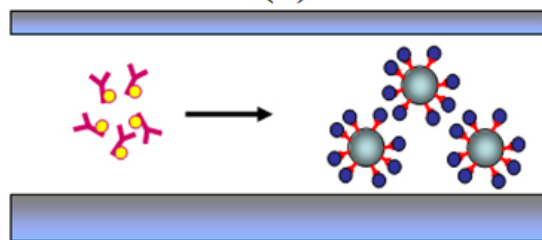
Microfluidic Version



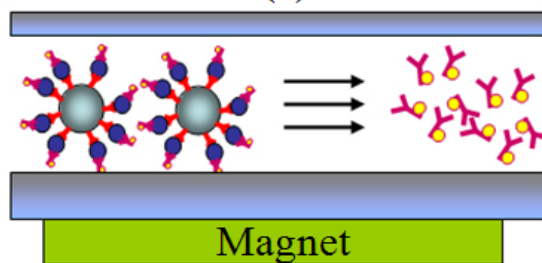
(a)



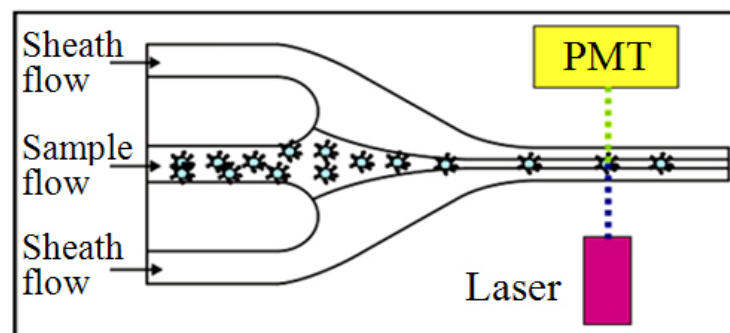
(b)



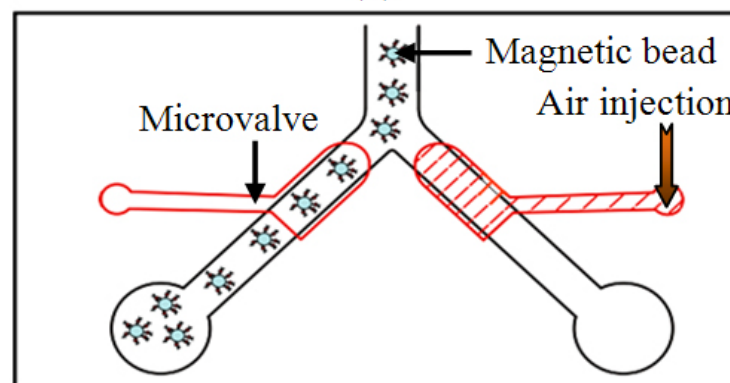
(c)



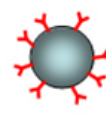
(d)



(e)



(f)



Capture antibody conjugated with a magnetic bead



Target virus

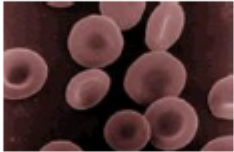
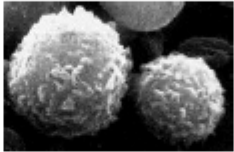
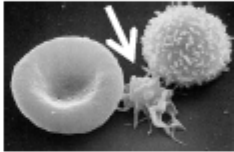
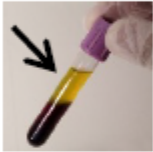


Custom developed antibody



The Case of Blood

Interest in extracting DNA from cells in solution:

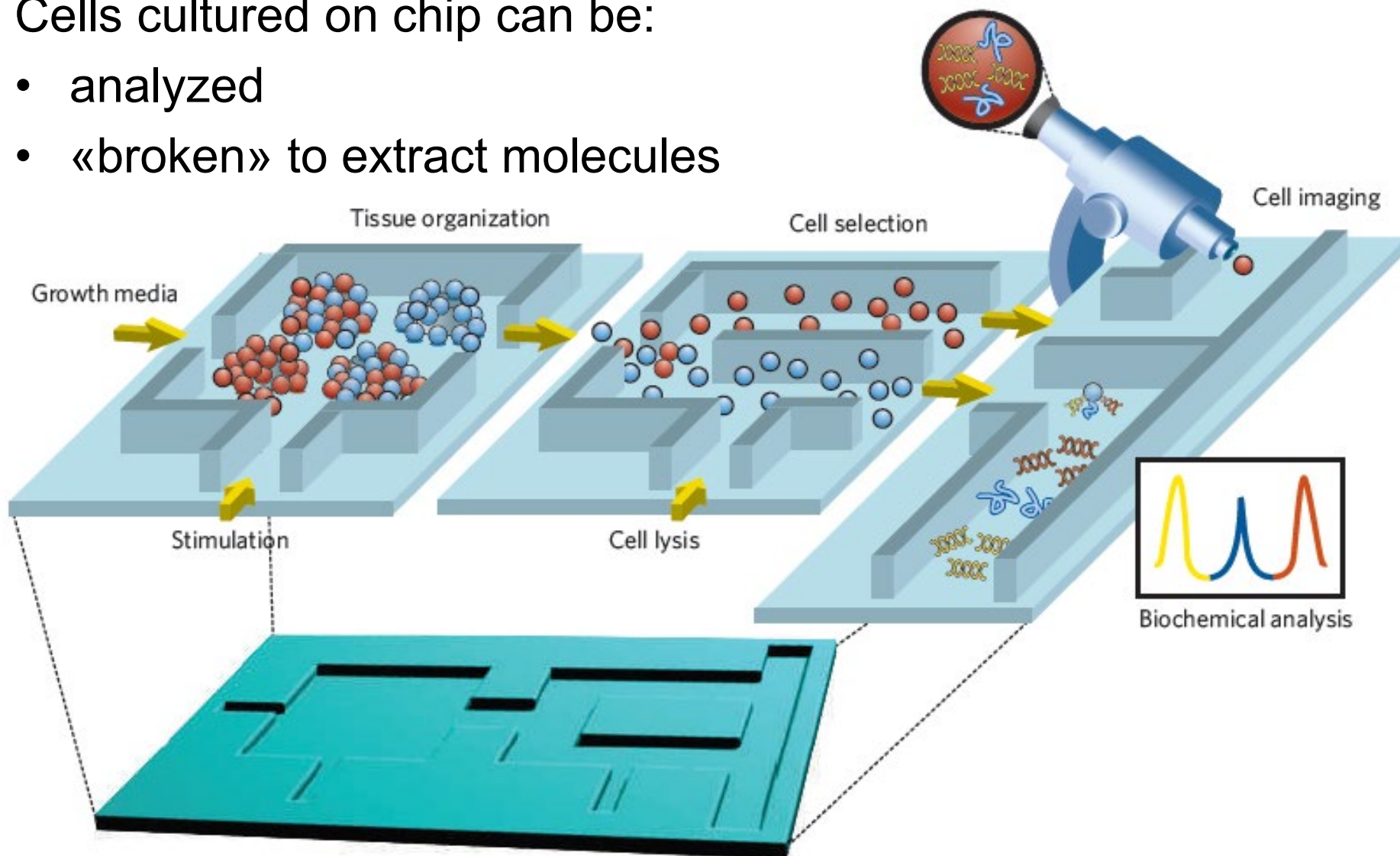
Components:	Bacteria	Erythrocytes	Leukocytes	Thrombocytes	Plasma
Common name	-	Red Blood Cells	White Blood Cells	Platelets	-
Amount / mL	$1 \cdot 10^4$	5×10^9	10^7	5×10^5	0,55 mL
Image					
Shape	Spheric, rod etc.	Disc shaped	Spherical	Irregular	Viscous fluid
Size	1-3 μm	6 x 6 x 2 μm	7-21 μm	2-3 μm	-
Nucleus	No	No	Yes	No	-
DNA	Yes	No (normally)	Yes	No	Random debris
Wall	Peptoglucan layer	Cell membrane	Cell membrane	Cell membrane	-
Surface antigens	Lipopolysaccharide, teichoic acids	HLA, CD239	HLA, CD45	HLA, GPIIb	Albumins, immunoglobulins



Fates of Cells on Chips

Cells cultured on chip can be:

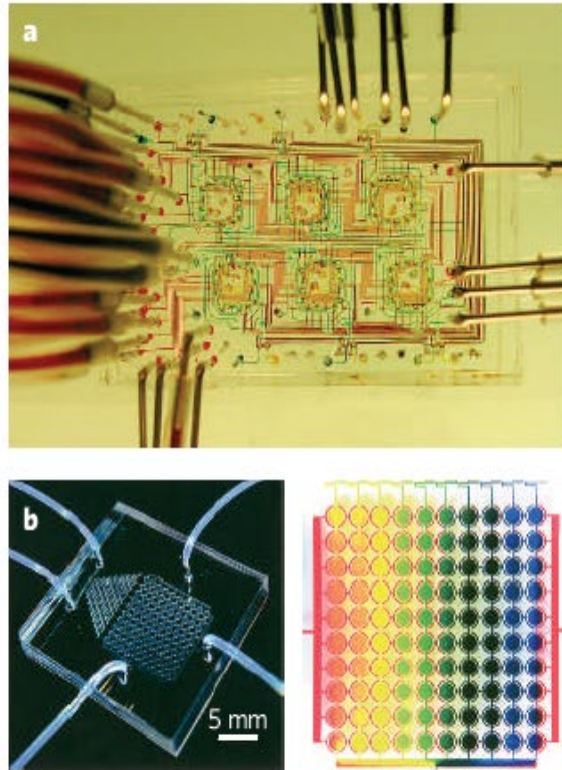
- analyzed
- «broken» to extract molecules



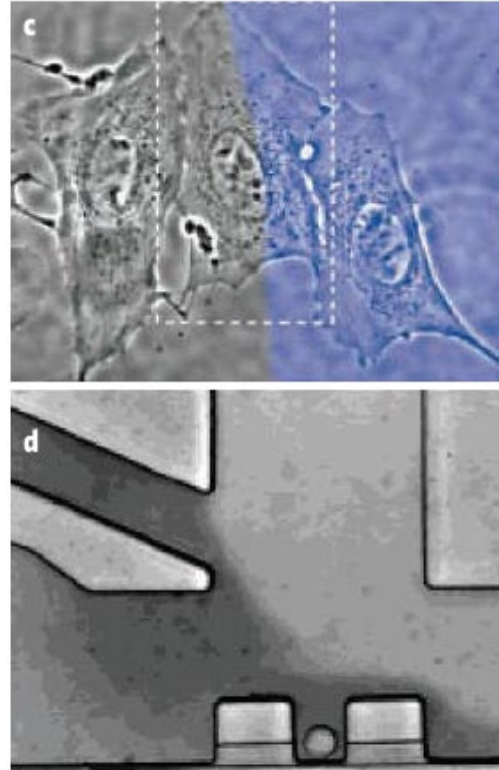


All the Steps Available in Microfluidics

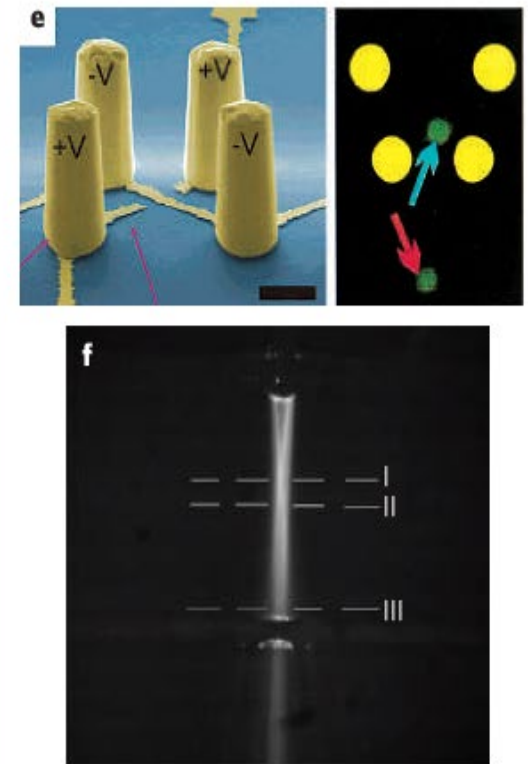
Culture



Treatment



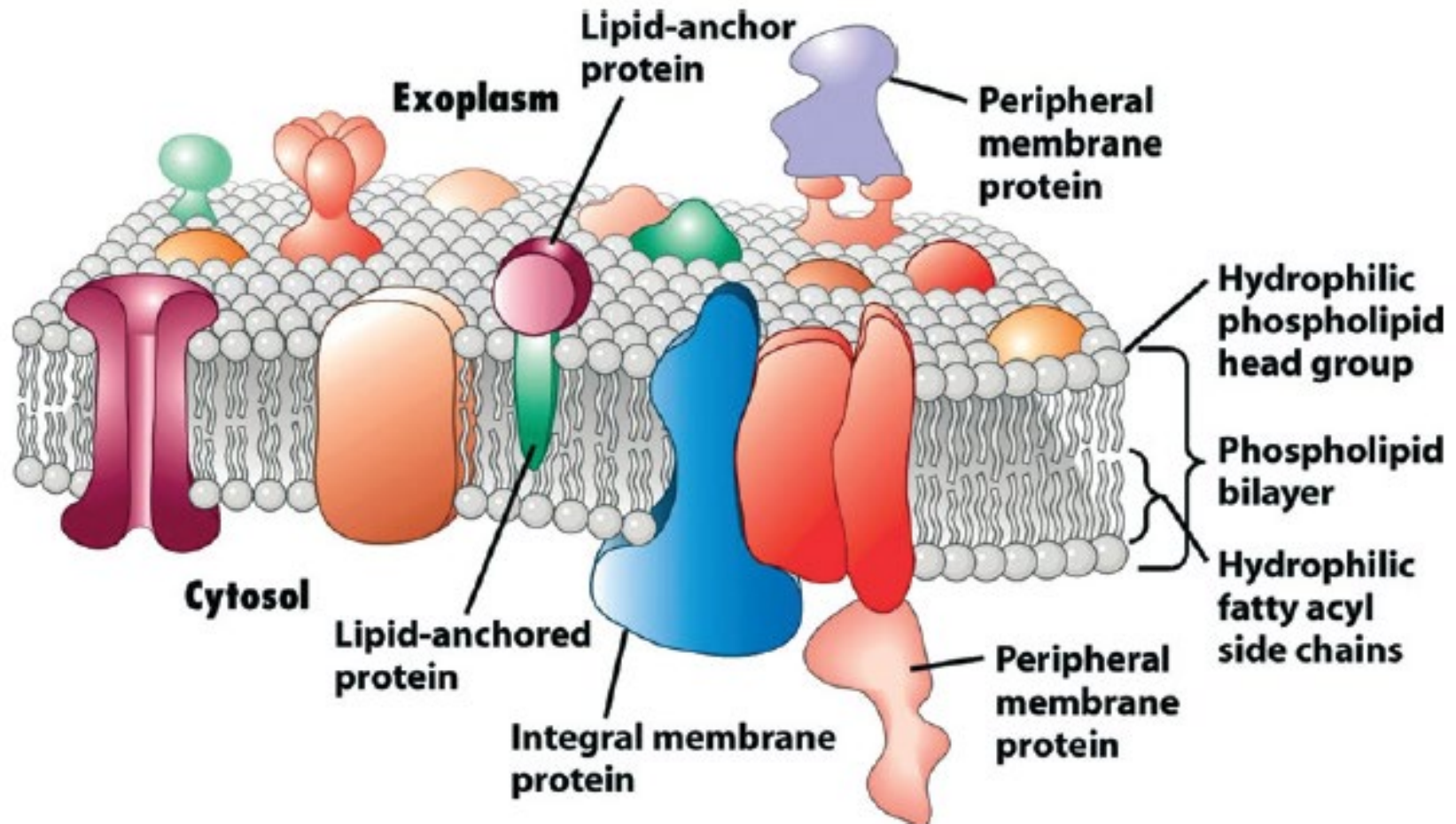
Selection





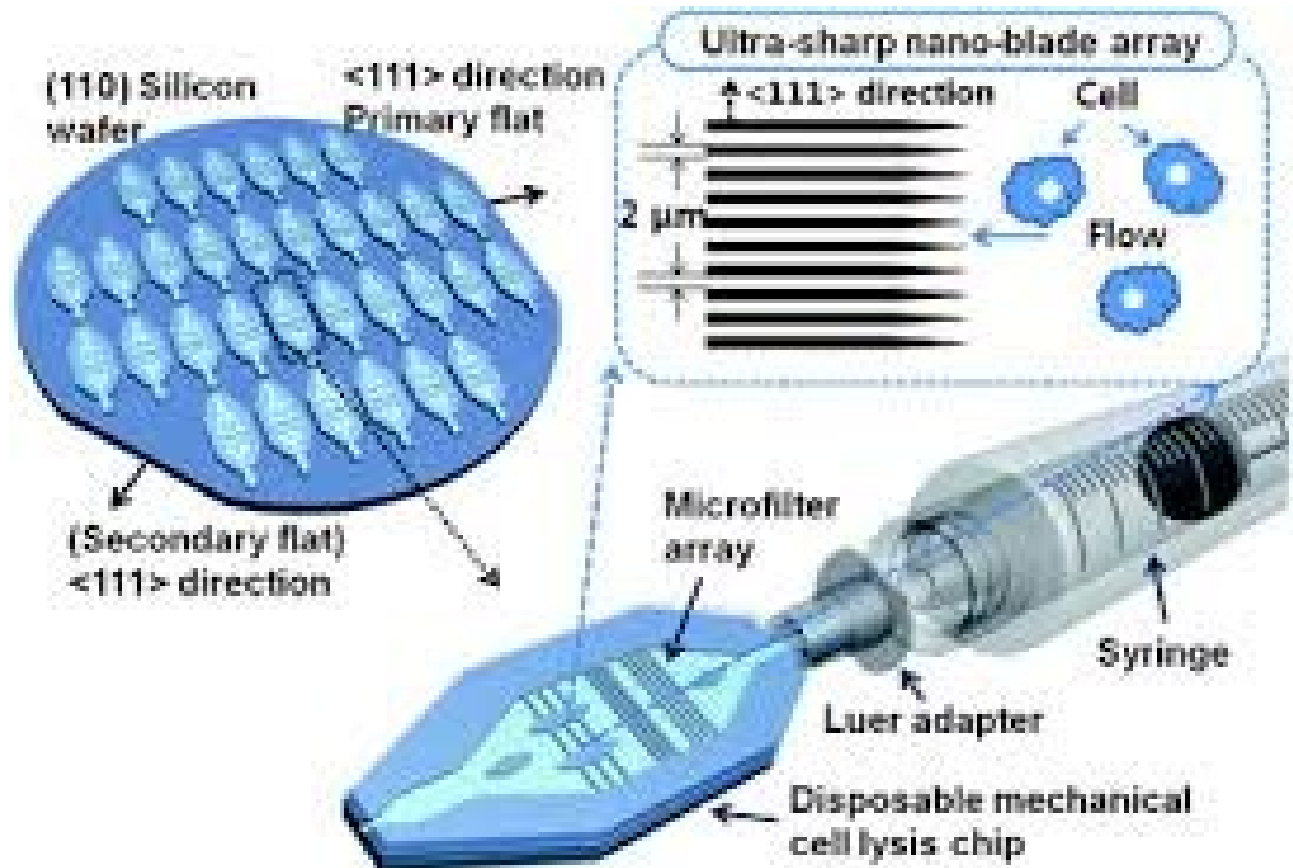
Cell Lysis: Cell Membrane

Very complex and robust: 4nm thickness, **dynamical**



Major approaches:

- Chemical
- Mechanical
- Electroporation
- Thermal

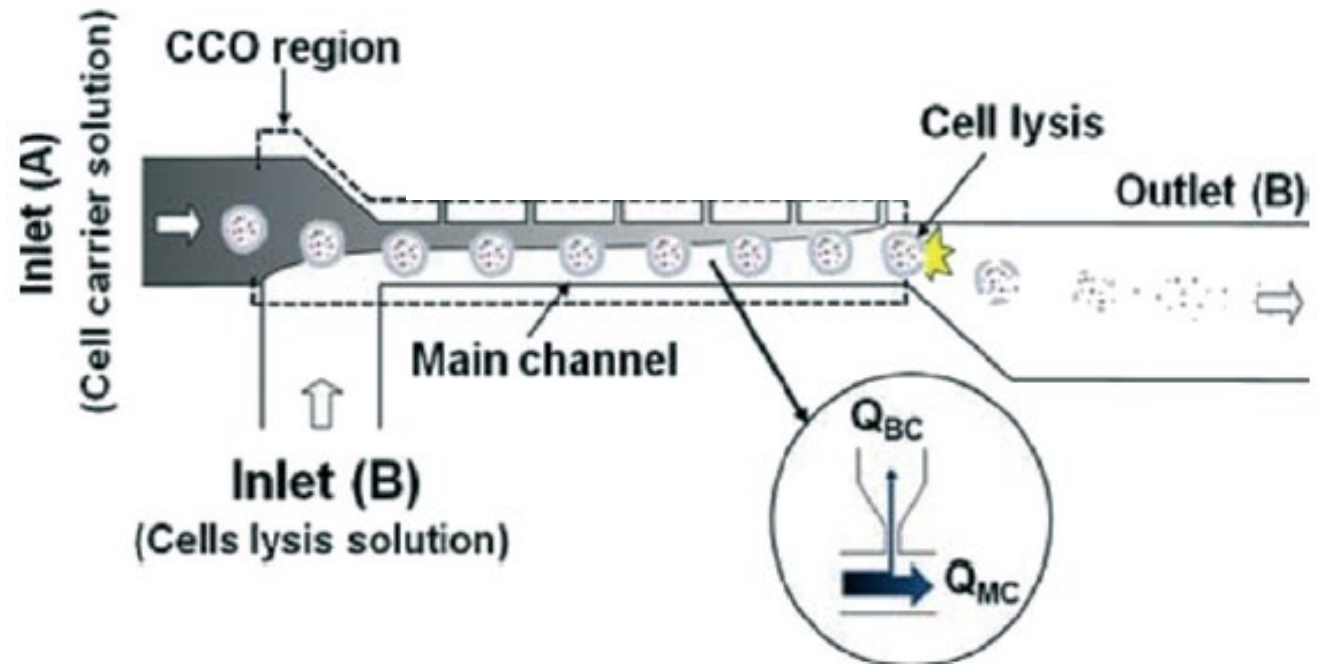




Chemical Lysis

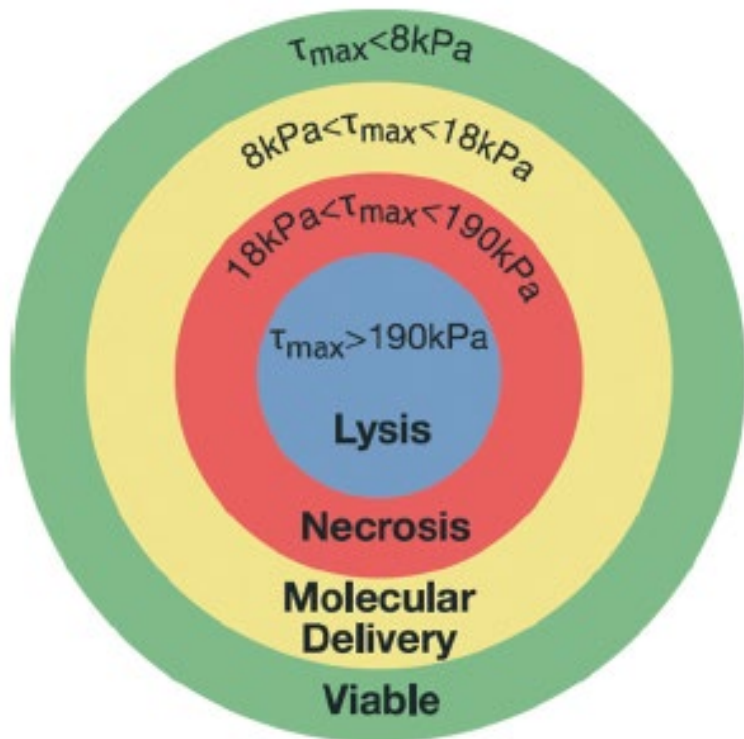
Chemical solutions with surfactants that solubilize the proteins:

- proteinase K
- sodium dodecil sulfate
- Triton X-100

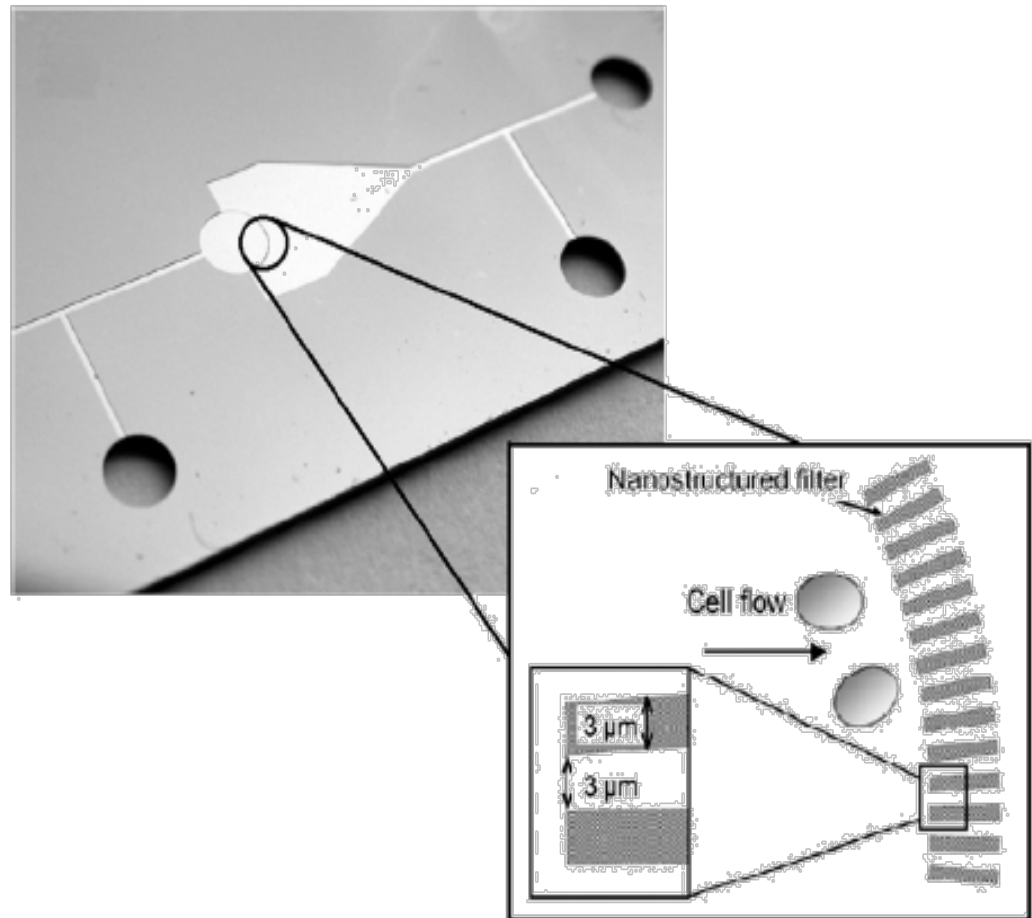


Mechanical: Impact Disrupture

Mechanical stress until rupture



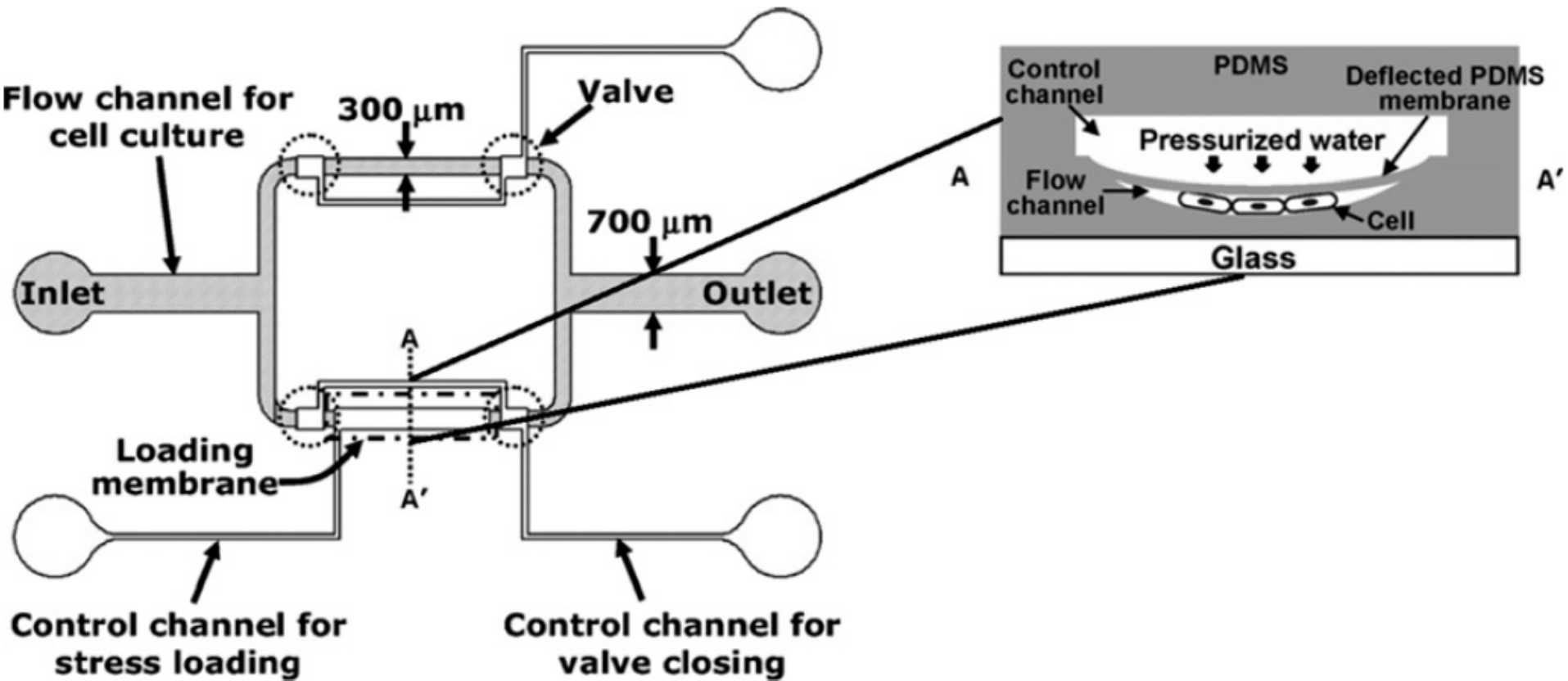
Impacts against sharp obstacles





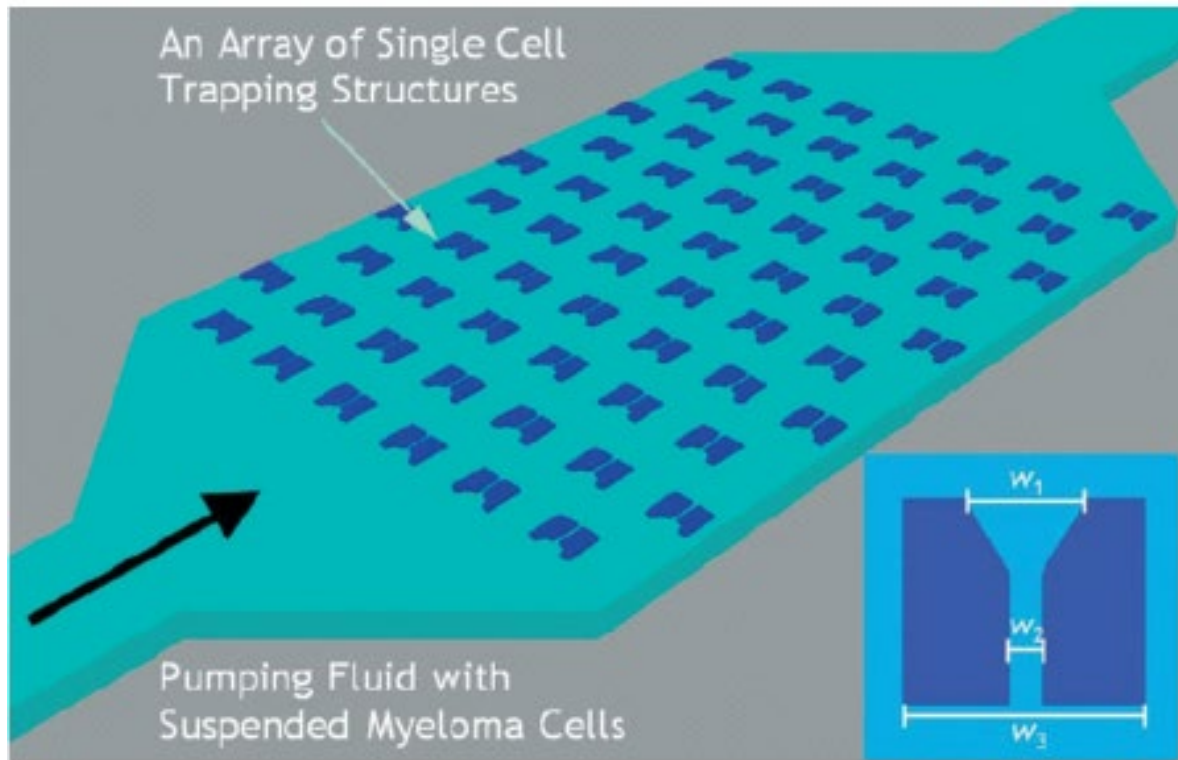
Compression

Lysis achieved at 35 kPa



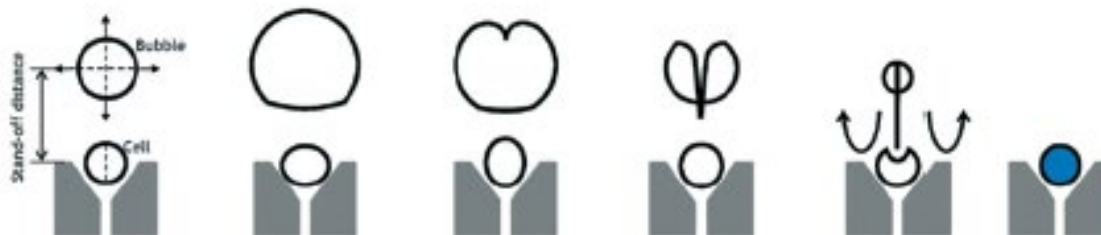
Other Mechanical Approaches

Air bubbles bombing



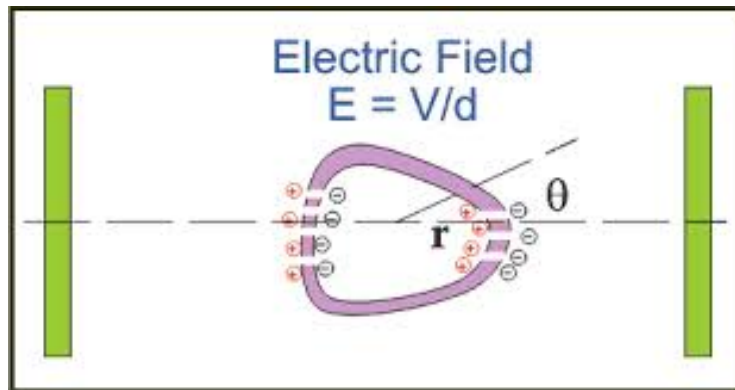
- Laser
- Magnetic beads
- Surface acoustic waves

cell trap

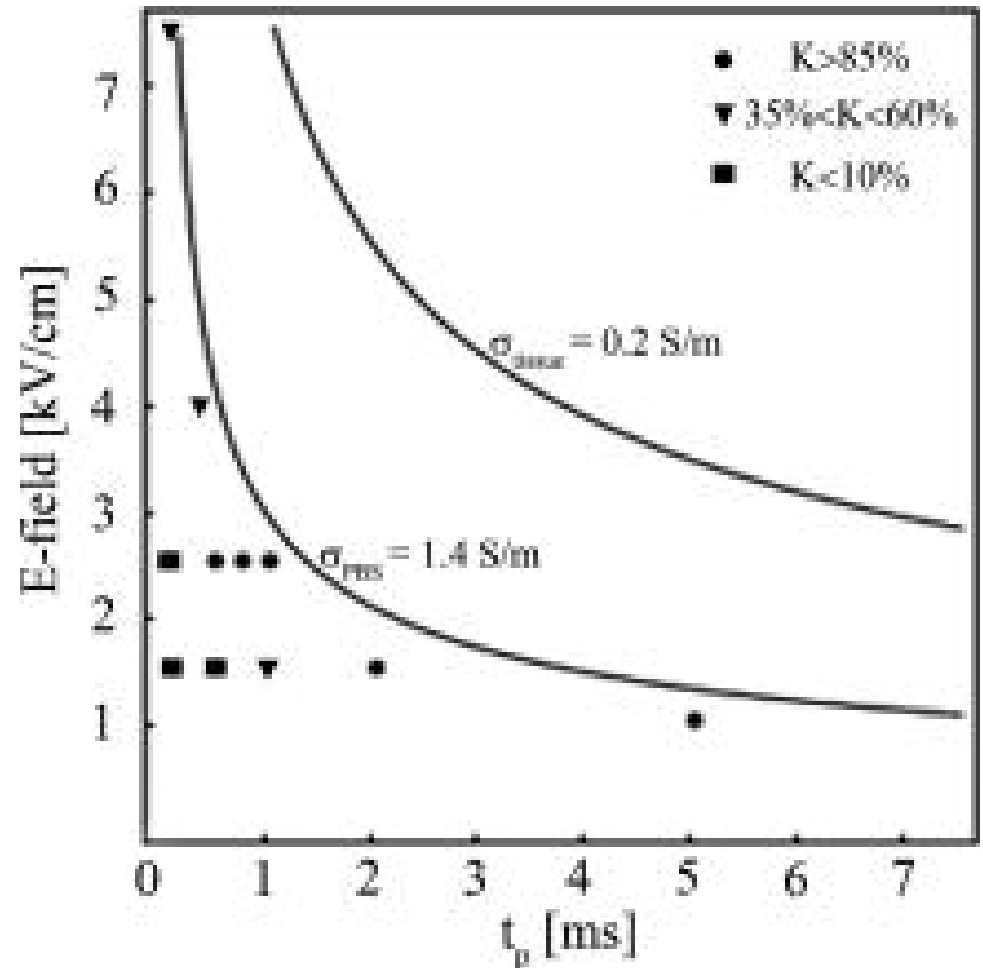


Electroporation: Field Intensity

Opening pores in the membrane with pulsated electric field

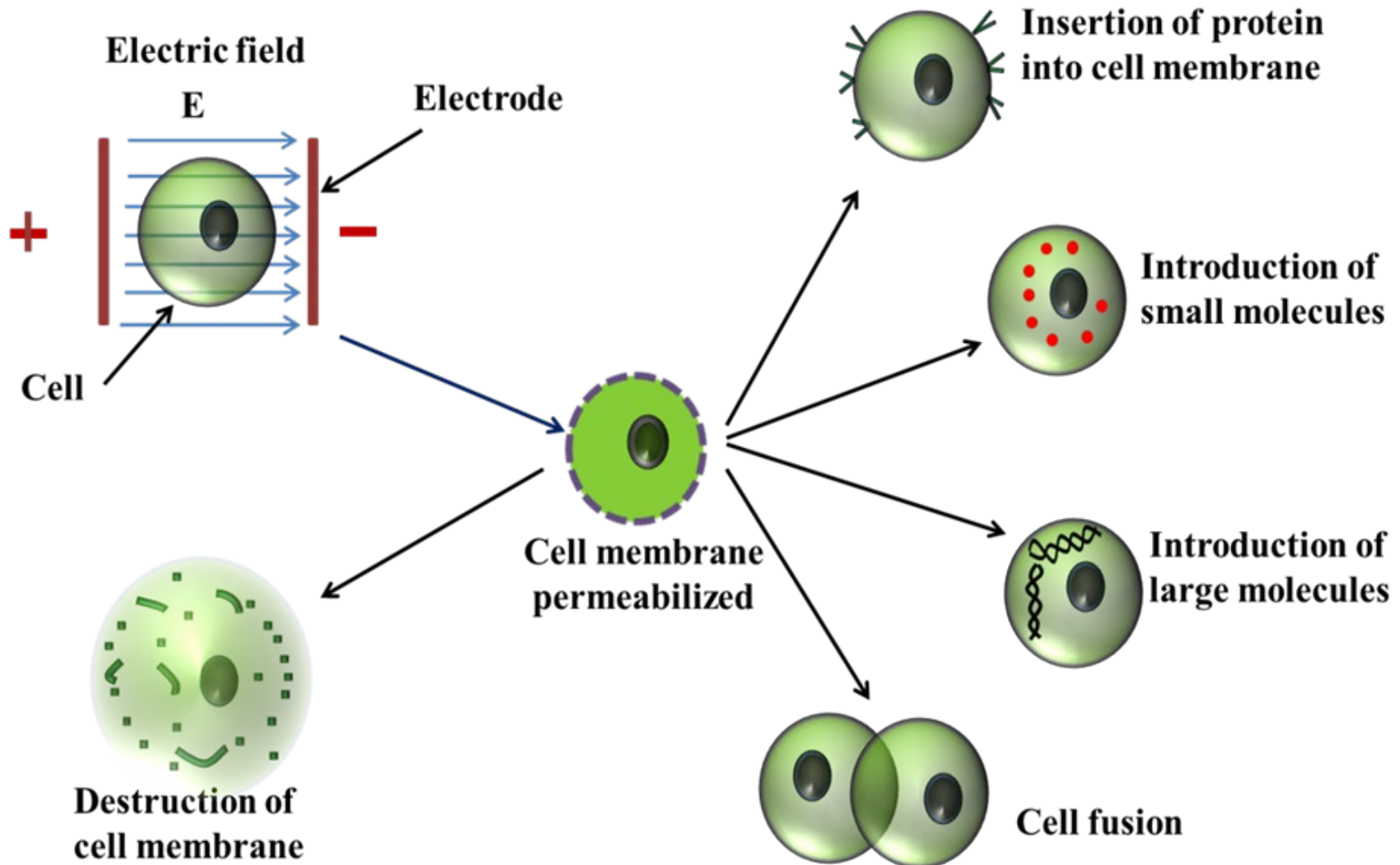


Threshold:
~1V across the
membrane



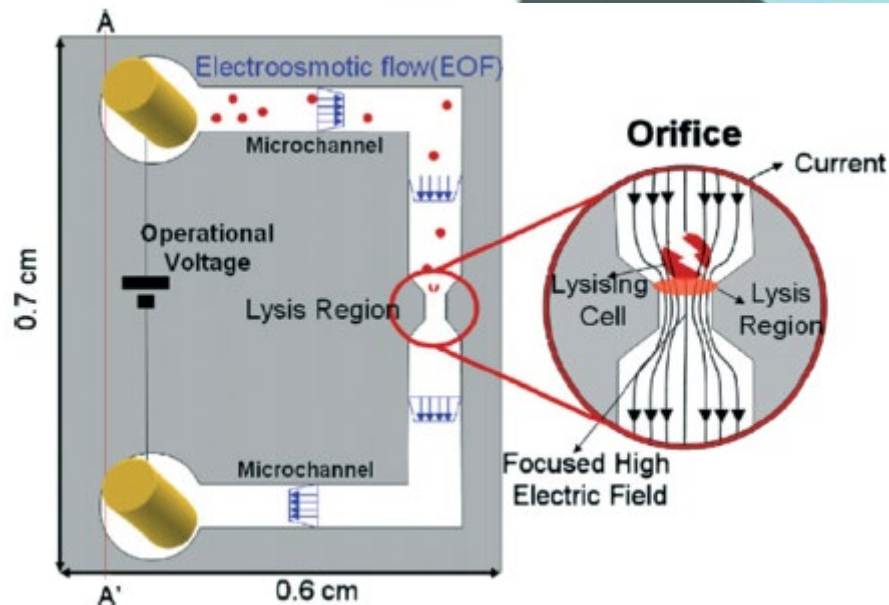
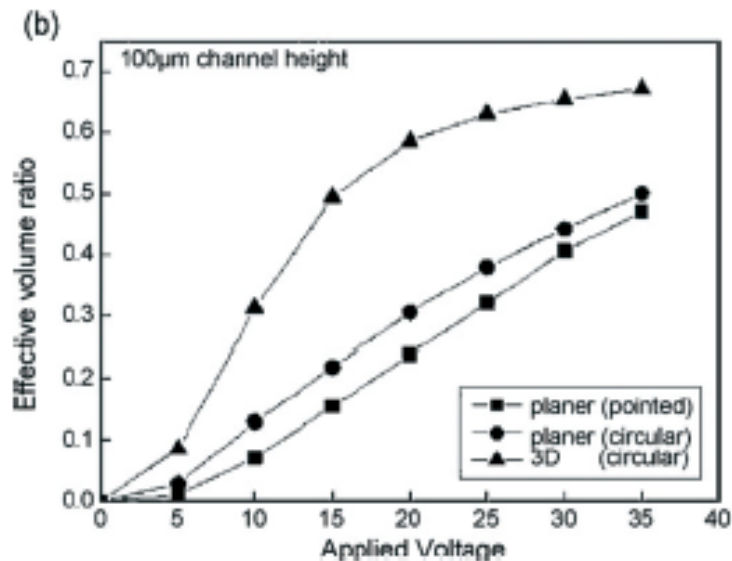
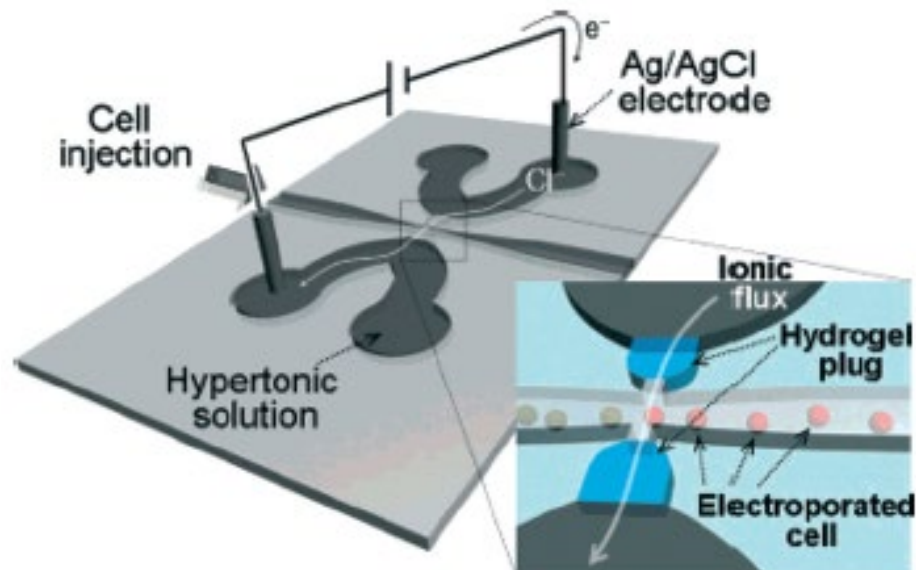
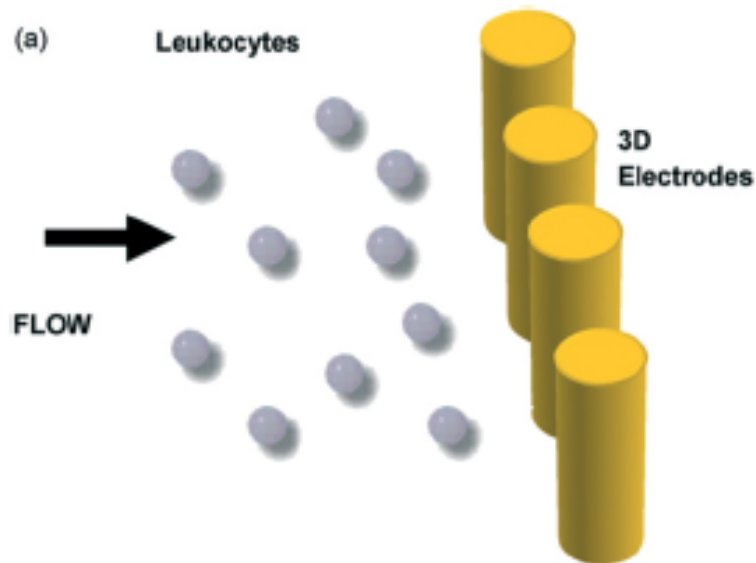
Electroporation: Applications

Used in different applications



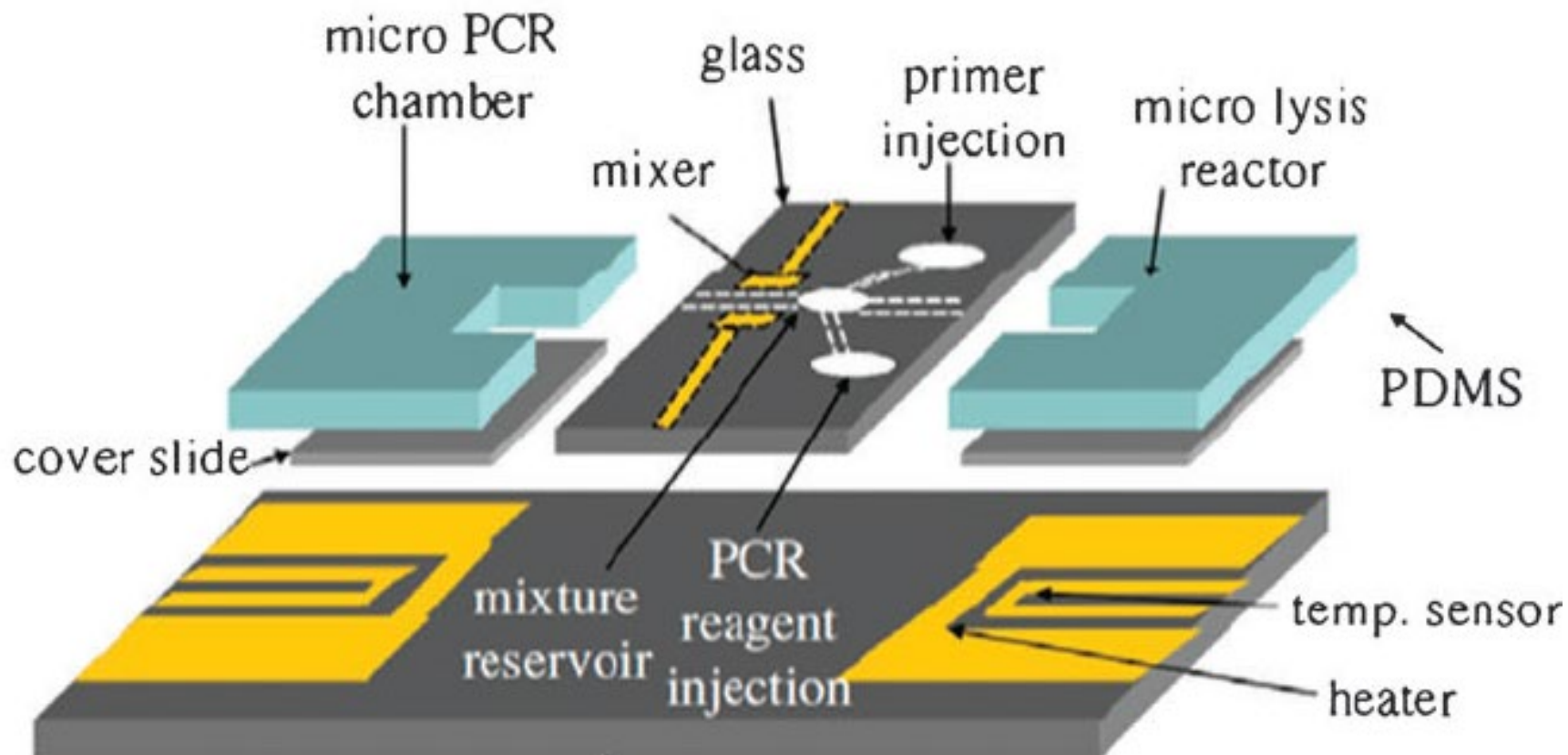


Different Implementations (DC & AC)





Temperature denaturates proteins (2 min at 95°C)





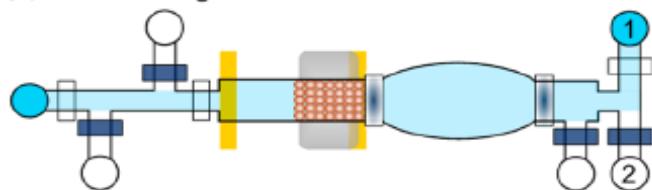
Comparison

Method	Fabrication techniques	Required controls	Lysis time	Comments/considerations
Chemical	Laser cut laminates, soft lithography (with SU8)	Flow control, temperature controls, valves	30–190 s	Reagent costs can be considerable. Chemistry needs to be modified for different cell types.
Electrical	Lithography, electrode deposition, glass bonding, electroplating, soft lithography	Voltage/current control	< 50 ms–20 min	Lower electric fields cause electroporation rather than lysis. Easily used with electrophoresis. Time for lysis depends on field strength.
Mechanical	DRIE, microfabrication, soft lithography	Flow control, Optical controls (laser), servo motors, pneumatic controls, ultrasonics	30 s–10 min	Good adaptability to different cell types. More involved fabrication and device operation.
Thermal	Microfabrication	Temperature controls, flow controls	2–5 min	Lysis can be sped up by increasing the pressure to prevent boiling at higher temperatures.

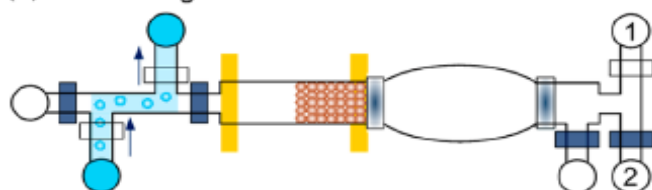
Lysis method	Efficiency	Lysing percentage	Technical difficulty	Cost	Brief comments
Chemical	+	++++	+	+	Simple and convenient, reagent residue must be removed.
Mechanical	++	+++	+++	++	Harmless to proteins, additional operation required.
Electroporation	++++	++++	+++	+++	Most widely used, accurate regulation of voltage required.
Laser	+++++	+++	++++	+++++	Highly efficient, complex and expensive system.
Thermal	++	++++	++	++	One of the earliest methods, causes damage to proteins.

Example of a Complete System

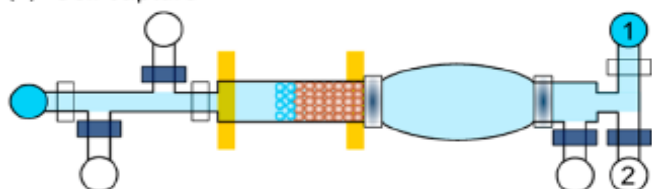
(a) Bead loading



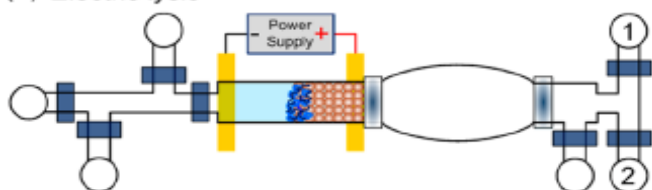
(b) Cell loading



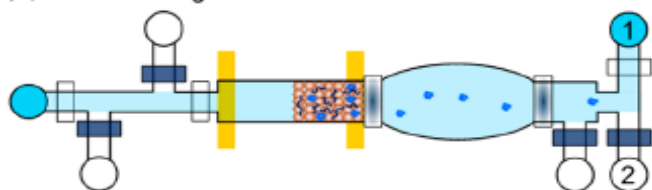
(c) Cell capture



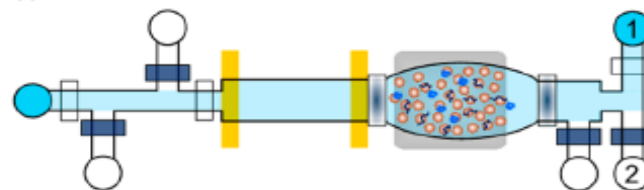
(d) Electric lysis



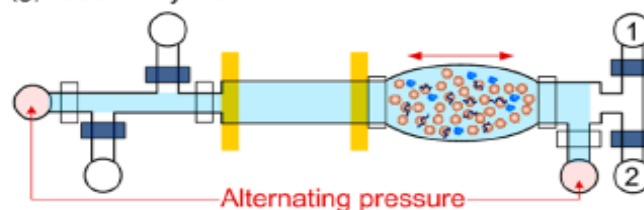
(e) DNA binding



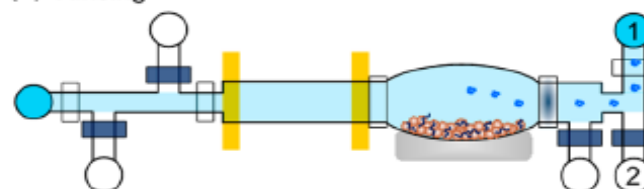
(f) Material transfer



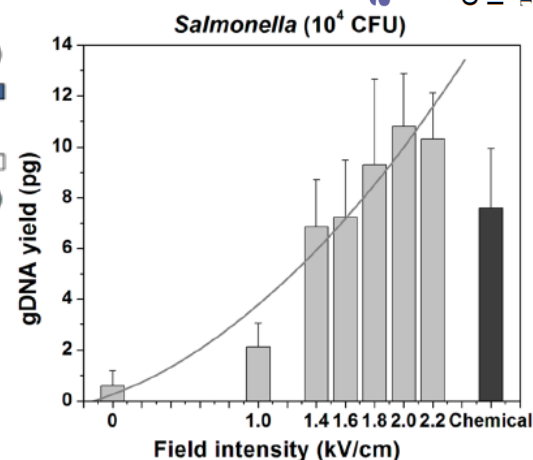
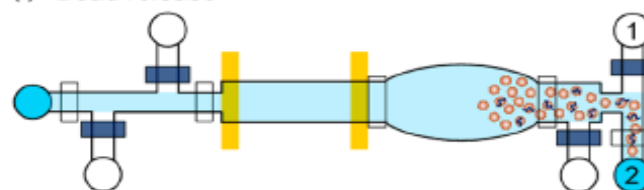
(g) Oscillatory movement



(h) Rinsing



(i) Bead release



analytical chemistry

Genomic DNA Extraction from Cells by Electroporation on an Integrated Microfluidic Platform
Tao Geng,[†] Ning Bao,^{‡,§} Nannalwar Sriranganathan,^{||} Liwu Li,[†] and Chang Lu^{†,§,||}



NATURE|Vol 442|27 July 2006|doi:10.1038/nature05063

INSIGHT REVIEW

Cells on chips

Jamil El-Ali¹, Peter K. Sorger² & Klavs F. Jensen¹

CRITICAL REVIEW

www.rsc.org/ibiology | Integrative Biology

Microfluidic sample preparation: cell lysis and nucleic acid purification

Jungkyu Kim,^{†a} Michael Johnson,^{†b} Parker Hill^a and Bruce K. Gale^{*b}

Received 24th March 2009, Accepted 10th August 2009

First published as an Advance Article on the web 25th August 2009

DOI: 10.1039/b905844c



Lab on a Chip

CRITICAL REVIEW

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Emerging microfluidic devices for cell lysis: a review

Lang Nan, Zhuangde Jiang and Xueyong Wei^{*}

Cite this: *Lab Chip*, 2014, 14, 1060